ASSIGNMENT – 2

CSA16 – DATA WAREHOUSING AND DATA MINING

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Q1 – MATCHING BUSINESS PROBLEMS TO MINING FUNCTIONALITIES (5 Marks)

TeleNova should select the mining functionality according to the business question. Different problems require different algorithms and types of knowledge.

Problem	Functionality / learning	Required data and source	Candidate algorithms and reason
P1: Plans/add-ons purchased together	Association; unsupervised. Produces frequent itemsets and rules such as plan A → add-on B.	Subscriber/order ID, plan, add-on, date; billing/warehouse transaction records.	Apriori and FP-Growth. Both find frequent itemsets and association rules; FP-Growth avoids repeated candidate generation.
P2: Predict customer leaving next month	Classification/prediction; supervised. Produces churn class or churn probability.	Historical churn label, usage, recharge, complaints, dropped calls, plan, tenure; CRM and usage warehouse.	Decision Tree and Naive Bayes. Tree gives understandable rules; Naive Bayes is a fast probabilistic baseline.
P3: Natural subscriber usage segments	Clustering; unsupervised. Produces groups with similar behaviour and no predefined labels.	Monthly usage, call minutes, recharge, data volume, tenure, add-on count; subscriber warehouse.	K-means and hierarchical clustering. K-means is efficient for large numeric data; hierarchical clustering shows nested groups.
P4: Unusual SIM calling behaviour	Outlier analysis; usually unsupervised. Produces unusual records for investigation.	Call frequency, destination diversity, call duration, time-of-day, roaming and location changes; call-detail records.	LOF and distance-based outlier detection. LOF compares local density; distance methods identify records far from normal behaviour.
P5: Characteristics of high- vs low-value	Concept/class description; descriptive	Revenue, recharge, tenure, usage, plan,	Decision Tree and OLAP/data-cube

customers	and may be supervised if value groups are labelled. Produces profiles and contrasts.	complaints and customer value class; CRM/warehouse.	summarisation. Trees reveal distinguishing rules; summaries compare group profiles.
P6: Monthly usage changes over 24 months	Evolution/trend analysis; descriptive/time-series analysis, generally not supervised classification. Produces trends, seasonality and changes.	Customer ID, month, data usage, call minutes and plan changes; historical warehouse.	Time-series trend analysis and regression. Trend analysis identifies direction/seasonality; regression estimates a usage trend.

The learning type depends on whether a known target label is available. Association, clustering and outlier detection generally work without labels; churn classification requires historical examples labelled as churn or non-churn.

Evaluation of the all-purpose algorithm proposal

The proposal is not suitable. Association rules discover co-purchase relationships, classification predicts a known class, clustering discovers unknown groups, outlier analysis detects rare behaviour, and evolution analysis studies changes over time. One algorithm cannot be expected to optimise all these different objectives. TeleNova should maintain a toolbox of algorithms and select them based on data type, scale, accuracy, interpretability and business purpose.

Q2 – THE KDD PROCESS, SYSTEM ARCHITECTURE AND MINING ISSUES (5 Marks)

2.1 KDD steps for MediCore

KDD step	MediCore activity
1. Selection	Select patients with completed hospital discharge records and identify those readmitted within 30 days. Collect admission, laboratory, pharmacy and billing attributes.
2. Preprocessing	Remove duplicate patient records, correct inconsistent codes, handle missing lab values,

	check impossible dates and reconcile patient identifiers.
3. Transformation	Combine records using patient/admission IDs; derive age, length of stay, medication count and previous-admission count; encode categorical variables and scale numeric values.
4. Data mining	Apply classification algorithms to discover patterns associated with 30-day readmission and estimate risk.
5. Interpretation/evaluation	Evaluate using recall, precision, F1-score and confusion matrix; review important patterns with clinicians and check whether findings are clinically meaningful.

2.2 Typical data mining architecture

Hospital sources: Admission | Laboratory | Pharmacy | Billing

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Database / Warehouse Server

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Cleaning, Integration, Transformation

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Data Mining Engine

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Pattern Evaluation Module ← Knowledge Base

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User Interface / Visualisation

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Clinicians and Hospital Analysts

The database/warehouse server stores integrated patient data. The mining engine applies classification and other algorithms. The knowledge base stores domain rules, metadata and thresholds. The pattern-evaluation module ranks useful patterns and removes uninteresting ones. The user interface allows analysts and clinicians to select data, run mining tasks and view results.

2.3 Classification of the MediCore system

View	Classification
Kinds of data	Relational and warehouse data: patient demographics, admission records, laboratory results, medicines and billing; time-stamped clinical records.
Kinds of knowledge	Classification rules, readmission prediction, association patterns and patient profiles.
Techniques used	Decision trees, Naive Bayes, logistic regression, clustering and statistical evaluation.
Applications	Clinical decision support, readmission prevention, resource planning and quality-of-care monitoring.

2.4 Data mining issues and mitigation

Issue category	Scenario-specific issue	Mitigation
Methodology	Readmission may depend on interacting clinical and social factors.	Use domain-guided feature selection, compare models and validate patterns with clinicians.
User interaction	Clinicians may not understand complex model output.	Provide simple explanations, risk factors, dashboards and feedback options.
Efficiency/scalability	Large multi-hospital records and repeated lab observations increase processing cost.	Use indexed warehouse tables, incremental ETL, sampling for exploration and scalable processing.
Data diversity	Admission, lab, pharmacy and billing data have different formats and units.	Use common patient/admission keys, standard codes, unit conversion and data validation.
Privacy/social	Patient information is sensitive; biased predictions may affect care.	Apply role-based access, encryption, de-identification, audit logs and fairness checks.

2.5 SQL query, statistics and data mining

A SQL query retrieves records that satisfy a condition, for example: SELECT patients WHERE length_of_stay > 7. A statistical summary may report the percentage of patients readmitted. Data mining goes further by discovering combinations of factors that predict readmission, such as repeated admissions, abnormal test results and particular medication patterns. It can generate a model that estimates risk for a new patient. Therefore SQL and statistics are useful, but they do not replace the pattern discovery and predictive modelling provided by data mining.

Q3 – DESCRIBING AND COMPARING DATA DISPERSION (7 Marks)

The data are already arranged in ascending order. For quartiles, use the median-of-halves method. For variance and standard deviation, treat the 24 recorded orders as the population being described (divide by N).

Measure	**Calculation / result**
Mean	$\Sigma x / N = 1143 / 24 = 47.625$ minutes
Median	Average of 12th and 13th values = (43 + 45) / 2 = 44 minutes
Mode	35 and 42 minutes (both occur twice; bimodal)
Midrange	(minimum + maximum) / 2 = (28 + 95) / 2 = 61.5 minutes
Range	95 − 28 = 67 minutes
Q1	Median of first 12 values = (36 + 38) / 2 = 37 minutes
Q3	Median of last 12 values = (53 + 55) / 2 = 54 minutes
IQR	Q3 − Q1 = 54 − 37 = 17 minutes
Population variance	$\Sigma(x - \text{mean})^2 / 24 = 213.32$ min²
Population standard deviation	$\sqrt{213.32} = 14.61$ minutes

Mean (47.625) is greater than median (44), suggesting a positively/right-skewed distribution. The high value 95 minutes pulls the mean upward.

3.1 Five-number summary and outlier test

Five-number summary: minimum = 28, Q1 = 37, median = 44, Q3 = 54, maximum = 95.

Lower fence = Q1 − 1.5(IQR) = 37 − 25.5 = 11.5. Upper fence = Q3 + 1.5(IQR) = 54 + 25.5 = 79.5. Since 95 > 79.5, 95 minutes is an upper outlier. No lower outlier exists.

Boxplot description: draw a box from 37 to 54, place the median line at 44, extend the lower whisker to 28 and the upper whisker to 72, and plot 95 separately as an outlier.

3.2 Recalculation after removing 95

Measure	All 24 values	After removing 95
Mean	47.625	45.565
Median	44	43
Mode	35 and 42	35 and 42
Midrange	61.5	50
Range	67	44
Q1	37	36
Q3	54	52.5
IQR	17	16.5
Population variance	213.32	120.77
Population standard deviation	14.61	10.99

The largest changes occur in midrange, range, variance and standard deviation because these measures are sensitive to extreme values. The median and quartiles change only slightly; the mode does not change.

3.3 Classification of dispersion measures

Measure	Category	Reason
Mean	Algebraic	Can be computed from partial counts and sums.
Median	Holistic	Exact median generally requires the full distribution or an order

		summary.
Variance	Algebraic	Can be combined from count, sum and sum of squares (or equivalent summaries).
Standard deviation	Algebraic	Derived from variance.
Range	Distributive	Partial minimum and maximum can be combined to obtain the global range.

Distributive measures can be computed from partitions and combined directly. Algebraic measures can be derived from a fixed number of distributive summaries. Holistic measures may require the complete data or a potentially large summary.

3.4 Tool verification and recommendation

The calculations can be verified in Excel using AVERAGE, MEDIAN, MODE.MULT, MIN, MAX, QUARTILE.INC, VAR.P and STDEV.P, or in Python using NumPy. Differences may occur if a tool uses sample variance (divide by $N-1$) or a different quartile convention. State the convention to make results comparable.

For a customer-facing typical delivery promise, recommend the median (44 minutes) because it is less affected by the 95-minute outlier. Also report the 90th percentile or an upper service-time measure and the IQR (17 minutes), so customers understand variability and slower deliveries.

Q4 – CLEANING NOISY AND INCOMPLETE DATA (6 Marks)

4.1 Defects and handling

Meter	Defect and type	Suggested handling
M102	kWh missing – missing value	Estimate using same meter's nearby valid hourly readings or class-wise mean for similar household/time. Do not use a global mean if local context is available.
M103	Region 'north' differs in case –	Standardise to 'North'.

	inconsistency	
M104	kWh = −14 – invalid value/noise	Check meter fault and source. Flag as invalid and exclude from analysis until corrected; do not replace blindly.
M105	Household type missing – missing value	Infer from customer/meter master; if unavailable use most probable value based on verified account information, otherwise mark Unknown.
M105	Voltage = 2300 V – likely unit/measurement error or extreme noise	Validate against device range and raw source. Quarantine for correction; do not smooth as an ordinary value.
M106	Region 'S.' – inconsistent abbreviation	Map approved abbreviation to 'South'.

The supplied 12 sorted kWh readings have mean = (5+9+11+14+18+19+23+26+27+30+33+38)/12 = 21.08 kWh. This can be used as a provisional global estimate only if no better meter/time or household-class information exists.

4.2 Equal-frequency binning (depth 4)

There are 12 sorted readings and 4 bins, so each bin contains 12/4 = 3 values.

Bin	**Values**	**Bin mean**	**Smoothed by mean**	**Smoothed by boundaries**
1	5, 9, 11	8.33	8.33, 8.33, 8.33	5, 11, 11
2	14, 18, 19	17.00	17, 17, 17	14, 19, 19
3	23, 26, 27	25.33	25.33, 25.33, 25.33	23, 27, 27
4	30, 33, 38	33.67	33.67, 33.67, 33.67	30, 30, 38

For boundary smoothing, each value is replaced by the nearest boundary. If distances are equal, use a consistent tie rule; here the lower boundary is selected for the value 33 in the last bin.

4.3 Regression and clustering for noise

- Regression: predict a sensor reading from related variables such as previous-hour reading, temperature and household type. Large residuals can be flagged as possible noise.
- Clustering: group normal readings with similar patterns. Values far from their cluster or outside the normal cluster range can be flagged for investigation.

A 2300 V reading is likely an invalid measurement or scaling/decimal error and must be checked against the meter's physical range. A 5 kWh reading may be a valid low-consumption observation and should not be removed merely because it is small.

4.4 Effect of global-mean imputation

Replacing all missing values with the global mean reduces measured variance because the inserted values lie at the centre of the distribution. It can weaken correlations, hide differences between household groups and create artificial clusters. Avoid it when data are missing for a reason, when groups have different typical values, or when missingness itself is informative. Use class-wise or model-based imputation and consider a missing-value indicator.

Q5 – INTEGRATION AND TRANSFORMATION OF DATA (7 Marks)

Part A – Integration

Conflict	Example	Resolution
Entity identification	Admissions student_id and LMS sid may identify the same student.	Use a master student key and a cross-reference mapping from both source IDs.
Naming	name versus full_name; scholarship versus scholarship_flag.	Adopt standard names such as student_name and scholarship_status.
Unit/format	dob may use different date formats; attendance may be stored as 0.8 or 80.	Convert dates to ISO date and attendance to percentage from 0 to 100.
Value conflict	hostel_status values may be Y/N while LMS uses true/false;	Create common controlled values and define an

	scholarship may disagree.	authoritative source or reconciliation rule.

Integration can create redundant attributes when two systems store the same fact under different names or derive one value from another. Correlation analysis can identify strongly related numeric attributes; domain knowledge is still required before removing one because correlation does not prove that two attributes are interchangeable.

Part B – Redundancy analysis

Pearson correlation for attendance {60,70,75,80,85,90} and marks {52,58,66,70,79,85}: $r = \Sigma[(x-\bar{x})(y-\bar{y})] / \sqrt{[\Sigma(x-\bar{x})^2 \Sigma(y-\bar{y})^2]} = 0.985$ (approximately). This indicates a very strong positive linear relationship: higher attendance is associated with higher marks. It does not prove causation.

Chi-square test for hostel status and scholarship:

Observed	**Scholarship**	**No scholarship**	**Total**
Hostel	120	180	300
Day scholar	80	320	400
Total	200	500	700

Expected counts	**Scholarship**	**No scholarship**
Hostel	300×200/700 = 85.71	300×500/700 = 214.29
Day scholar	400×200/700 = 114.29	400×500/700 = 285.71

$\chi^2 = \Sigma(O-E)^2/E = 33.60$. Degrees of freedom = (2−1)(2−1) = 1. Since 33.60 > critical value 3.84 at 0.05 significance, reject the null hypothesis of independence. Hostel status and scholarship are statistically associated.

Part C – Transformation

Annual family incomes: ₹20,000, ₹35,000, ₹50,000, ₹80,000, ₹150,000. Minimum = 20,000; maximum = 150,000; mean = 67,000; population standard deviation = 46,000.

Income (₹)	Min-max (0 to 1)	Z-score	Decimal scaling
20,000	0.0000	−1.0217	0.020
35,000	0.1154	−0.6957	0.035
50,000	0.2308	−0.3696	0.050
80,000	0.4615	0.2826	0.080
150,000	1.0000	1.8043	0.150

Min-max formula: $(x-20{,}000)/(150{,}000-20{,}000)$. Z-score formula: $(x-67{,}000)/46{,}000$. Decimal scaling: divide by 10^6 so the largest absolute value is below 1.

Euclidean distance before normalisation for (19, 35,000) and (21, 80,000): $\sqrt{[(21-19)^2 + (80{,}000-35{,}000)^2]} = 45{,}000.000$ (approximately).

After min-max normalisation: age values use range 17–25 and income uses range ₹20,000–₹150,000. Normalised differences are 2/8 = 0.25 and 45,000/130,000 = 0.34615. Distance = $\sqrt{(0.25^2 + 0.34615^2)} = 0.427$ (approximately).

The original distance is dominated by income because its numeric scale is much larger. Normalisation puts attributes on comparable scales, which is important for distance-based algorithms such as K-means and nearest-neighbour methods.

- Aggregation: calculate average marks for each department or semester.
- Generalisation: replace individual dates of birth with age groups such as 18–20 and 21–23.
- Attribute construction: derive academic performance category from marks, or derive age from date of birth.

Q6 – DATA REDUCTION AND DISCRETIZATION (7 Marks)

Part A – Choosing reduction strategies

Category	Technique	Suitability for sensor data
Data-cube aggregation	Aggregate readings by machine, hour, day and sensor type.	Useful for dashboards and long-term trends; keep detailed rows for failure diagnosis.
Attribute subset selection	Forward selection, backward elimination or decision-tree induction.	Select sensors that contribute to failure detection and remove irrelevant/redundant attributes.
Dimensionality reduction	PCA.	Compress correlated sensor variables into components while retaining most variation; retain original critical sensors where interpretability is required.
Numerosity reduction	Sampling, histograms or clustering.	Summarise millions of readings; retain rare failure cases separately.
Compression	Lossless compression or controlled lossy compression.	Reduce storage and transfer cost. Use lossless for exact event records; lossy only when it does not erase failure signals.

Sampling method	Meaning	Assessment
Simple random without replacement	Each row can be selected once.	Avoids duplicates but may miss rare failures.
Simple random with replacement	A row may be selected more than once.	Can duplicate records and still under-represent rare failures.
Cluster sampling	Select complete groups such as machines or time blocks.	Efficient when groups are representative, but selected clusters may be biased.
Stratified sampling	Split into classes/groups and	Recommended:

	sample within each.	oversample/retain the 0.5% failure class and sample normal readings separately.

If only 0.5% of rows are failures, use stratified sampling so the training set contains enough failure examples. Preserve a separate test set with the real-world class distribution. Do not evaluate on an artificially balanced test set.

Keep the smallest number of principal components that explains a chosen proportion, for example 95% of training-data variance, then validate failure recall. PCA must be fitted on training data only. If the 95% threshold removes important failure information, retain more components or selected original sensors.

Part B – Discretization by calculation

Equal-width: minimum = 3, maximum = 44, width = (44−3)/4 = 10.25.

Interval	**Values assigned**	**Count**
[3, 13.25)	3, 7, 7, 9, 12	5
[13.25, 23.5)	14, 14, 15, 18, 21, 21, 22	7
[23.5, 33.75)	25, 25, 27, 31	4
[33.75, 44]	36, 36, 40, 44	4

Equal-frequency: 20 values ÷ 4 bins = 5 values per bin.

Bin	**Values**	**Count**
1	3, 7, 7, 9, 12	5
2	14, 14, 15, 18, 21	5
3	21, 22, 25, 25, 27	5
4	31, 36, 36, 40, 44	5

Entropy-based split

There are 5 Y and 5 N labels, so parent entropy H(S) = 1 bit. For each candidate midpoint, information gain = $H(S) - (nL/10)H(L) - (nR/10)H(R)$.

Split midpoint	Left labels (Y,N)	Right labels (Y,N)	Information gain (bits)
11.0	0,4	5,1	0.1080
13.5	0,4	5,1	0.2365
16.5	0,4	5,1	0.3958
20.0	0,4	5,1	0.6100
23.5	1,4	4,1	0.2781
26.5	2,4	3,1	0.1245
29.5	2,5	3,0	0.3958
33.0	2,6	3,0	0.2365
37.5	2,6	3,0	0.1080

The best split is at 20.0, between values 18 (N) and 22 (Y), with information gain approximately 0.6100 bits.

Equal-width and equal-frequency discretization are unsupervised because they use the values and their distribution, not class labels. Entropy-based discretization is supervised because it uses the known class labels. It is more suitable for classification when the split is validated and does not overfit.

Q7 – DIAGNOSE, REDESIGN AND VALIDATE (13 Marks)

A. Diagnose the computational bottleneck (2 Marks)

Original pipeline decision	Problem / principle violated	Effect
Replace missing income with global mean	Poor missing-data treatment; ignores skew and groups.	Pulls values toward centre, reduces variance and hides income differences.
Drop occupation with 18% missing	Premature attribute removal; ignores potentially useful information.	Loses predictive information and may introduce bias if missingness is systematic.
Five equal-width income bins	Unsuitable discretization for highly right-skewed data.	Most observations may fall in a few bins while high incomes occupy sparse bins.
Normalise before train/test split	Data leakage; preprocessing must be learned from training data only.	Test-set minimum/maximum influence training transformation and make evaluation optimistic.
Random 10% training sample; only 3% defaults	Severe class imbalance and inadequate minority representation.	Training may contain too few defaulters; model learns majority class and misses defaults.
97% accuracy	Wrong metric for imbalanced data.	A classifier predicting every loan as non-default can achieve about 97% accuracy while detecting no defaulters.

B. Measurable design objectives (2 Marks)

- Primary objective: improve recall for the default class, because missing defaulters causes financial loss.
- Report precision, recall, F1-score, PR-AUC, ROC-AUC, confusion matrix and accuracy. Use recall as the primary metric and monitor precision to control false alarms.
- Use a stratified train/validation/test split and keep the test set at the natural class distribution.

- Constraints: no test-data leakage; retain useful occupation information; handle missing values consistently; keep preprocessing reproducible and explainable; satisfy available compute and time limits.

C. Revised preprocessing strategy (4 Marks)

Order	Revised step	Justification and fit rule
1	Inspect, validate and integrate data; remove duplicate loan IDs and correct invalid values.	Prevents source errors. Use documented business rules; not a learned transformation.
2	Split into stratified train, validation and test sets before fitting preprocessing.	Preserves minority examples and prevents leakage. Test set remains untouched.
3	Handle missing income using training-set median, optionally by relevant group; add an income-missing indicator.	Median is robust to right skew; fit median on training data only. Prefer over global mean.
4	Retain occupation; impute using most frequent category or an explicit 'Unknown' category and add a missing indicator.	Avoids discarding potentially predictive information. Learn imputation categories from training data only.
5	Treat skewed income using log1p or robust scaling; inspect outliers before transformation.	Log transformation reduces right skew. Fit scaling parameters on training data only; compare with equal-frequency bins if discretization is needed.
6	Use supervised or quantile discretization only if validation improves performance.	Equal-frequency handles skew better than equal-width; entropy split uses labels. Fit bin boundaries on training data only.
7	Apply feature selection only within training/cross-validation folds.	Reduces irrelevant variables without using test information.

8	Handle class imbalance with class weights or training-only resampling such as SMOTE.	Preserves test distribution and improves minority learning. Resampling is performed only on training folds.
9	Fit normalisation/encoding on training data and apply unchanged to validation/test.	Prevents leakage and ensures consistent deployment processing.
10	Train and tune a baseline and revised classifier; select threshold using validation data.	Optimise default recall subject to an acceptable precision/false-positive cost.

D. Predict before implementing (1 Mark)

Prediction: log transformation or robust scaling will reduce the influence of very high incomes and make the income distribution less right-skewed. Stratified splitting and training-only class balancing will increase the proportion or effective weight of defaults seen during training, while the test distribution remains unchanged. Accuracy may decrease slightly as the model identifies more defaulters, but default recall should improve; precision may decrease because more customers are flagged.

E. Implement and validate (2 Marks)

Use a public credit dataset such as the UCI German Credit dataset or a similar approved dataset. Create two pipelines: (1) reproduce the original preprocessing decisions as closely as possible; (2) implement the revised leakage-free pipeline. Use the same stratified test set for both. In each pipeline, fit imputers, encoders, scalers, feature selection and resampling using training data only.

Metric	**Original pipeline**	**Redesigned pipeline**
Accuracy	Enter measured result	Enter measured result
Default-class precision	Enter measured result	Enter measured result
Default-class recall	Enter measured result	Enter measured result
Default-class F1-score	Enter measured result	Enter measured result
PR-AUC / ROC-AUC	Enter measured result	Enter measured result
Confusion matrix	Attach screenshot/output	Attach screenshot/output

A result must be measured by running the experiment. Do not fill the table with assumed accuracy or recall. Attach dataset source, preprocessing settings, code/workflow, random seed and screenshots. For a fair comparison, both pipelines must use the same train/test split and test rows.

F. Defend the final design (2 Marks)

Choose the redesigned pipeline if it improves default recall and PR-AUC on the untouched test set while maintaining an acceptable precision and operational false-positive rate. Compare the observed results with the predictions in Part D. A remaining risk is data drift: customer behaviour and economic conditions can change, reducing model performance. Monitor recall, precision, default rate, missing-value rates and feature distributions after deployment, and retrain only through a governed validation process.

CONCLUSION

The assignment demonstrates how data mining functionalities are selected for different business problems and how KDD converts raw data into useful knowledge. Data cleaning, integration, transformation, reduction and discretization improve data quality and mining efficiency. Dispersion and statistical tests help describe data and detect relationships. The loan-default case shows why leakage-free preprocessing, minority-class handling and suitable evaluation metrics are essential.

SUBMISSION EVIDENCE CHECKLIST

- Show manual calculations for Q3, Q4, Q5 and Q6.
- Attach a boxplot for delivery times and identify the 95-minute outlier.
- Verify calculations using Excel, Python, Weka or KNIME screenshots.
- Attach the Q7 dataset source, code/workflow, settings and confusion matrices.
- Fill Q7 comparison table using actual measured results; do not submit placeholder values as results.